
Higher-Order Drug-Drug Interaction Prediction

Technical Review & Architecture Evaluation

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What is Polypharmacy & The Clinical Problem

Clinical Context & Statistics

Definition: Polypharmacy is the simultaneous, routine administration of multiple medications by a single patient, typically defined clinically as taking 5 or more drugs concurrently.

- **Prevalence:** Over 40% of older adults take 5+ prescription medications, and nearly 20% take 10 or more simultaneously.
- **Impact:** Adverse Drug Events (ADEs) result in over 1.3 million emergency department visits and 350,000 hospitalizations annually in the US alone.
- **Risk:** There is an exponential increase in unpredictable, synergistic adverse events as the size of the drug regimen grows, complicating manual pharmacological review.

Project History & Computational Gap

Historically, pharmacovigilance datasets (like **SIDER** and **OFFSIDES**) focused entirely on single-drug side effects. Later efforts, such as **TWOSIDES**, expanded this scope but were strictly limited to pairwise (2-drug) interactions.

The Gap: Real-world prescriptions are *higher-order sets* (3, 4, 8+ drugs). Standard graph networks (like Decagon) rely on clique expansion algorithms that destroy high-order group fidelity by forcing multi-drug sets into isolated two-node connections. Our goal is to predict specific side effects directly from the **entire** intact multi-drug set.

Dataset Engineering: FAERS to HODDI

Our base data is derived from the **FDA Adverse Event Reporting System (FAERS)**, spanning 41 quarters (2014Q3 to 2024Q3). We transformed noisy observational records into the standardized Higher-Order Drug-Drug Interaction (HODDI) benchmark through a rigorous extraction pipeline.

1. Side Effect Normalization

Raw, unstructured FAERS complaint strings were embedded via **SapBERT** and mapped to standard MedDRA terms. These were converted to stable UMLS CUIs. We applied a strict confidence gate (cosine similarity ≥ 0.9) to discard ambiguous text and isolate verified side effects.

2. Drug & Structural Encoding

Noisy drug trade names were resolved to explicit **DrugBank IDs** (e.g., "WARFARIN SODIUM" mapped to DB00682).

IDs were then mapped to **SMILES** notation strings. These strings were parsed into character-level tokens to feed our CNN for automated chemical feature extraction.

3. Evaluation & Data Scale

- **Final Scale:** 109,744 records, comprising 2,506 unique drugs and 3,950 unique SEs.
- **Filtering:** Superset removal was applied to ensure only parsimonious (minimal necessary) combinations were kept.
- **Validation:** Strict chronological split—Train (70% 2014-2021), Val (15% 2022), Test (15% 2023-2024).

HGNN-SA Model

- 🔗 Hypergraph convolution retains multi-drug groupings.
- 📄 SMILES CNN extracts 64-dim structural fingerprint.
- 🔍 Set-attention pooling isolates specific toxicity drivers.

HGNN_train.py (Line 38)

```
x = F.relu(self.conv1(node_feats, edge_idx)) x =  
F.relu(self.conv2(x, edge_idx)) # Set Attention Pooling  
drugs = x[edge_idx[0]] attn_probs = scatter_softmax(  
self.attn_layer(drugs), edge_idx[1], dim=0 )  
combo_vector = scatter_add( drugs * attn_probs,  
edge_idx[1], dim=0 )
```

HGNN Results

Source: evaluate_HGNN.txt | Scheme A Labels

0.890
ROC-AUC

0.899
PR-AUC

0.805
F1-SCORE

0.818
ACCURACY

0.749
RECALL

	Predicted Safe (0)	Predicted Toxic (1)
Actual Safe (0)	11,730	1,483
Actual Toxic (1)	3,304	9,909

PolyFormer Model

-  Fully inductive set transformer architecture.
-  Zero global graph adjacency matrix required.
-  Side effect actively queries the specific drug set.

polyformer_train.py (Line 92)

```
drug_ctx = self.encoder( drug_feat,  
src_key_padding_mask=pad_mask ) q =  
self.se_emb(se_idx).unsqueeze(1) attn_out, _ =  
self.cross_attn( query=q, key=drug_ctx, value=drug_ctx,  
key_padding_mask=pad_mask ) combo = attn_out.squeeze(1)
```

PolyFormer Results

Source: Evaluate_PolyFormer.txt | Scheme A Labels

0.967

ROC-AUC

0.967

PR-AUC

0.894

F1-SCORE

0.898

ACCURACY

0.853

RECALL

	Predicted Safe (0)	Predicted Toxic (1)
Actual Safe (0)	12,425	737
Actual Toxic (1)	1,927	11,235

SECTION 2

Negative Label Stress Tests

The Negative Sampling Flaw (Scheme A)

Standard pharmacovigilance databases (like FAERS) only report positive interactions. Synthetic negative generation is required for supervised learning. **Scheme A (the official HODDI method)** creates negatives by simultaneously corrupting both a drug and the side effect.

Double-Corruption Algorithm

For every true positive $\{A, B, C\} \rightarrow SE^*$:

1. Replace one random drug (e.g., $C \rightarrow D$)
2. Replace the side effect (e.g., $SE^* \rightarrow SE'$)
3. Yield candidate $\{A, B, D\} \rightarrow SE'$ as negative.

Metric Inflation

By corrupting **both axes simultaneously**, the algorithm creates obvious structural artifacts. The model learns to detect the distributional shift of the corruption pattern rather than underlying pharmacology. This explains the highly optimistic **0.967 ROC-AUC**—the task is artificially easy.

The Subset False-Negative Bug

A critical flaw in standard negative sampling occurs when a generated negative candidate still contains a smaller, known toxic subset. This results in labeling a genuinely toxic combination as "safe" (a false negative), severely hindering the model's ability to learn true synergistic effects.

Clinical Reality (Ground Truth)

The drug pair **{A, B}** synergistically causes side effect **SE***.

FAERS reports a patient taking **{A, B, C}** experiencing **SE***. This is recorded as a true positive.

Flawed Negative Generation

The algorithm selects the positive **{A, B, C}** and corrupts drug C into D.

The new candidate is **{A, B, D}**.

Result: {A, B, D} is labeled NEGATIVE (-1), despite containing the toxic {A, B} subset known to cause SE*.

STRESS TEST 01

Scheme B: Drug-Corrupt Only

To test exact drug membership, we replace one drug while **keeping the side effect fixed**. This removes the SE-distribution cue, making the task significantly harder.

Implementation snippet:

```
# generate_synthetic_negatives.py for _ in range(MAX_ATTEMPTS): idx =
rng.randrange(len(drugs)) if drugs else 0 current = set(drugs) candidates
= [d for d in drug_pool if d not in current] new_drug =
rng.choice(candidates) new_drugs_list = list(drugs) new_drugs_list[idx] =
new_drug new_set = combo_key(list(dict.fromkeys(new_drugs_list))) key =
(new_set, se) if key in positive_keys: continue # Retry if key collision
```

Test Protocol

- Dataset: hoddi_synth_B.csv
- Replaces exactly one drug.
- Target SE remains identical.
- Evaluates drug-set specificity.
- **Exposes the subset bug (~0.56% contamination).**

Scheme B Results

Source: Evaluate_PolyFormer_synth_B.txt

0.780

ROC-AUC

0.798

PR-AUC

0.655

F1-SCORE

0.553

RECALL

	Predicted Safe (0)	Predicted Toxic (1)
Actual Safe (0)	11,343	1,745
Actual Toxic (1)	5,850	7,238

STRESS TEST 02

Scheme C: SE-Corrupt Only

To test side-effect specificity, we keep the **exact drug combination** intact and only swap the side effect. This evaluates if the model knows *which* adverse event is caused by a specific polypharmacy set.

Implementation snippet:

```
def try_se_corrupt(drugs, se, se_pool, positive_keys, rng): drug_set =  
    combo_key(drugs) others = [s for s in se_pool if s != se] if not others:  
    return None for _ in range(MAX_ATTEMPTS): new_se = rng.choice(others) #  
    Verify the new combo+SE isn't a known true positive if (drug_set, new_se)  
    not in positive_keys: return new_se return None
```

Test Protocol

- Dataset: hoddi_synth_C.csv
- Drug set remains 100% identical.
- Randomizes the SE target.
- Evaluates multi-label discrimination.
- Immune to the subset contamination bug.

Scheme C Results

Source: Evaluate_PolyFormer_synth_C.txt

0.890

ROC-AUC

0.892

PR-AUC

0.815

F1-SCORE

0.815

RECALL

	Predicted Safe (0)	Predicted Toxic (1)
Actual Safe (0)	10,680	2,408
Actual Toxic (1)	2,415	10,673

Scheme D: Subset-Aware Generation

Scheme D resolves the subset false-negative bug found in Scheme B. While performing a drug-corrupt swap, we cross-reference the proposed candidate against a mapping of all known positive combinations for that specific side effect. If the candidate acts as a superset to any known toxic subset, it is rejected.

Subset Detection Function

```
def contains_positive_subset(candidate: FrozenSet[str], se: str,
se_to_combos: Dict[str, List[FrozenSet[str]]]) → bool: """True if any known
positive combo for this SE is a subset of candidate.""" for pos in
se_to_combos.get(se, []): if len(pos) ≤ len(candidate) and
pos.issubset(candidate): return True return False
```

Rejection Pipeline

```
# Inside the drug-swap generation loop (Scheme D) contaminated =
contains_positive_subset(new_set, se, se_to_combos) if scheme == "D" and
contaminated: local_subset_skips += 1 continue # Reject candidate and retry
found = (sorted(new_set), contaminated) break # Accept valid candidate
```

Scheme D Results

Source: Evaluate_PolyFormer_synth_D.txt

0.792

ROC-AUC

0.810

PR-AUC

0.659

F1-SCORE

0.546

RECALL

	Predicted Safe (0)	Predicted Toxic (1)
Actual Safe (0)	11,641	1,445
Actual Toxic (1)	5,933	7,155

Generation Statistics

Source: synthetic_negative_generation_stats.csv

Scheme	Positives	Negatives	Failed	Subset Contamination
B (Drug Corrupt)	111,072	111,072	0	~0.56%
C (SE Corrupt)	111,072	111,072	0	0% (N/A)
D (Subset Aware)	111,072	111,061	11	0%

ROC-AUC Comparison Across Schemes



PolyFormer Full Metric Comparison

Source: synthetic_negative_comparison.csv

Scheme	ROC-AUC	PR-AUC	F1-Score	Accuracy	Recall
A (HODDI Official)	0.9678	0.9672	0.8940	0.8988	0.8536
C (SE-Corrupt)	0.8905	0.8925	0.8157	0.8157	0.8155
D (Subset-Aware)	0.7923	0.8102	0.6598	0.7181	0.5467
B (Drug-Corrupt)	0.7802	0.7987	0.6559	0.7098	0.5530

Drivers of Success

Bypassed Reduction

Maintains higher-order native structures.

Avoids pairwise simplification noise.

Dynamic Attention

Actively learns specific drug drivers.

Avoids uniform group averaging.

Inductive Capability

Set transformer fully independent.

Zero global graph connectivity required.

Future Scope

- Interactive Streamlit web interface.
- Visualizing model attention weights in a bid to make it more explainable.

Questions & Discussion

Thank you for your time.

